

The role of AI in therapy

With how mental healthcare is generally experiencing an increasing amount of pressure, more effort is continuously being devoted to the idea of using AI as a tool that can help in creating diagnoses and assisting in providing good quality care. First, we describe the main ways that effort came to fruition.

1. Psychotherapy augmentation

The most empirically supported application of AI in the current literature appears to be AI-driven augmentation of human-based psychotherapeutic practice. Promising findings show that AI can augment practitioners' analyses of interviews and other client data through natural language processing (NLP; Beg et al., 2024). NLP models were also found to aid the identification of symptoms of depression and anxiety, allowing for tailored interventions based on individual data (Ke et al., 2025).

Beyond just language processing, AI also seems to be particularly useful in analyzing patterns. For instance, when applying machine learning methods instead of classical statistics for key diagnostic indicators, ML models were shown to accurately predict treatment outcomes and code linguistic data in 36 out of 37 analyzed studies (Aafjes-van Doorn et al., 2021). In general, AI has shown promising results when finding details which would not be oft spotted, including questionnaire analysis, EEG signals pointing to certain possible disorders, detection of stress through physiological data analysis and more (Abrar et al., 2026).

2. AI-based chatbots

AI-based chatbots like Woebot and Wysa have been shown to reduce symptoms of depression and anxiety by using CBT techniques (Nelson et al., 2025), with other models also showing more general improvements in user well-being, frequency of healthy behavior and psychoeducation (Beg et al., 2025). In fact, at least one study directly compared an AI-based chatbot with a conventional therapeutic intervention for anxiety and found that, although the model had weaker performance to that of humans (35% to 50% respectively), it still showed significant improvements (Spytska, 2025). Overall, this is a promising field of research likely to expand massively in the upcoming years as AI psychotherapy models become more refined and more direct experiments are conducted.

3. Accessibility and scalability

Lastly, nearly all of the literature mentioned above highlights how AI can be used to increase the scalability and accessibility of psychotherapy (e.g. Aafjes-van Doorn et al., 2020). By creating a low-cost, 24/7 available psychotherapeutic solution, the reach of therapy is likely to significantly expand, helping both practitioners with overbearing workloads, as well as clients by allowing them to participate in (perhaps inferior) therapy.

However, not everything is so rosy. Studies also highlight major limitations regarding technological and methodological weaknesses, as well as ethical and privacy concerns. The following paragraphs present some of the most pressing matters.

Technology & methodology	Ethics & privacy
Despite promising early findings, it is crucial to keep in mind that psychotherapeutic applications of AI are still significantly technologically limited. Most researchers warn against blind acceptance of AI output due to processing biases that occur when its training dataset is not fully representative (e.g. Ke et al., 2025). This problem is further compounded by the “black box” nature of most models, making the process of reaching their outputs impossible to track. Unfortunately, even research seems to poorly control for bias, with most studies having insufficient training data, questionable generalizability and improper use of machine learning terminology (Aafjes-van Doorn et al., 2020). Other findings include limitations of AI-based chatbots’ crisis intervention	Similar to accessibility advantages, ethics and privacy concerns are mentioned in nearly every paper (e.g. Ke et al., 2025). With the sensitivity of mental health data, it is paramount for AI models to be protected from breaches, unauthorized access and the misuse of user data (Abrar et al., 2026). Compliance with legal standards must be programmed into the models in order to ensure user safety. Likely, new international standards will need to be developed for therapeutic AI implementation and application (Abrar et al., 2026). Beyond legality, however, the technological limitations of AI also provide ethical challenges. For instance, systematic bias tends to negatively affect minority cultures due to disproportionate data distributions in training samples (Nelson et al., 2025). Mechanisms that detect and mitigate these biases are

(Nelson et al., 2025), clinicians' skepticism of accepting AI as augmentative to psychotherapy, infrastructure difficulties with adopting new models in outdated IT systems, and an overall lack of regulations and use guidelines (Abrar et al., 2026).

necessary. Similarly, the previously mentioned black box nature of AI makes for a lack of transparency, often crucial when diagnosing or conceptualizing client cases (Ke et al., 2025). Finally, there is a very real risk of overreliance on AI. It was already found that 69 out of a sample of 116 clinicians (59%!) used AI in their clinical practice (Linardon et al., 2025), and this pattern is unlikely to change as the technology becomes more advanced.

So, with all that being said, let us return to the main question from the beginning of this segment: "Can AI replace psychotherapists?" Although current studies show substantial support for AI augmenting human-led psychotherapy, beyond theoretical considerations, there is simply not enough research experimentally comparing AI to practitioners yet in order to reach a conclusion. Furthermore, before such research truly flourishes, major ethical, technological and theoretical limitations need to be overcome. Thus, it appears that, as of right now, we can only make educated guesses about the future.

You can imagine that this is a highly debated topic among experts (have a look at e.g. Brown et al., 2019). However, as the Website Team at PAIR, we propose an approach that takes a step back and focuses on the wider dynamics of nature vs technology as a whole. This perspective is heavily inspired by our psychotherapy professor and research lab supervisor Rafał Styła, after having a class discussion with him regarding psychotherapy and AI (personal communication, January, 2026). We sincerely thank him and his thorough professional expertise on the subject matter.

Final verdict

Considering certain ideas from evolutionary psychology, humans evolved from early hominids millions of years ago in an environment much different from the one we are familiar with today. Back then, archeological evidence suggests that these early humans

lived in tightly-knit communities, with large emphases on social interactions and relations. Although the rapid advances in technology have allowed us to move past physical barriers and become less interdependent through computers, the internet, and even AI, the evolution of human nature is far slower since it is based on genetic changes. Thus, our brains are adapted to the stone world of old, while living in the modern age of today.

Regarding our question, this means that the desire for human interaction is evolutionarily ingrained within us, and it can only change through a process of genetic adaptation that takes millions of years to complete. As long as we are aware that we are conversing with something non-human, helpful as it may be, an element of humanity will always be lacking. Thus, although AI-based psychotherapy methods that are equivalent or even superior to humans might very well emerge, **the human practice of psychotherapy seems unlikely to be fully replaced.** Simply put, our very nature requires it.

References

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